

Chapter 8: Our Skeletal System and the Skin

Textbook Exercise Solutions & Answers



1. Fill in the blanks with the proper word.

(a) The place where two or more bones are connected is called a

joint

(b) Cells of epidermis contain a pigment called

melanin

(c) and are the two layers of the human skin.

epidermis, dermis

(d) The human skeletal, system is divided into parts.

two

2. Match the pairs.

Column 'A' Column 'B' 1. Ball and socket joint a. Knee 2. Hinge joint b. Wrist 3. Gliding joint c. Shoulder

Column 'A'	Column 'B'
1. Ball and socket joint	c. Shoulder
2. Hinge joint	a. Knee
3. Gliding joint	b. Wrist

3. Right or wrong? If wrong, write the correct sentence.

(a) Bones are soft.

Wrong: Bones are hard.

(b) The human skeleton protects the internal organs.

Right

4. Put a [✓] mark at the proper places.

(a) The system which gives our body a definite shape to (a) Excretory system [] (b) Respiratory system [] (c) Skeletal system [] (d) Circulatory system []

(c) Skeletal system [✓]

(b) The joint is seen in fingers and toes. (a) Hinge joint [] (b) Ball and socket joint [] (c) Immovable joint [] (d) Gliding joint []

(a) Hinge joint [✓]

5. Answer the following questions in your words.

(a) What are the functions of your skin?

The functions of the skin are :

1. Protects the internal parts of the body like muscles, bones, organ systems etc.
2. Help to preserve the moisture in the body.
3. Synthesizing Vitamin D.
4. Regulates body temperature by releasing sweat.
5. Gives protection from heat and cold.
6. Functions as the sensory organ of touch

(b) What should you do to keep the bones strong and healthy?

To keep the bones strong and healthy:

1. We should include calcium and phosphorus rich food in our diet.
2. We also include vitamin D rich food in our diet.
3. We get these from milk, milk products, leafy vegetables, meat and exposure to sunlight.
4. We should exercise regularly.

(c) What are the functions of human skeletal system?

The functions of human skeletal system are:

1. Gives a definite shape to the body.
2. Provides support to the body.
3. Protects the delicate organs inside the body.

(d) Which are the various reasons due to which our bones might break?

The bones in our body might break due to:

1. Lack of calcium and phosphorous.
2. Due to fracture because of an accident or a fall.
3. Lack of proper nutrition.
4. Due to deficiency of Vitamin D.

(e) What are the different types of bones? How many types are there? Give example of each.

There are four types of bones in our body:

Types of Bone	Example
1. Flat bones	Sternum in the chest
2. Small bones	Stirrup in each ear
3. Irregular bones	Vertebra posterior (back side) of the body
4. Long bones	Femur or thigh bone

Types of Bone Example

1. Flat bones Sternum in the chest
2. Small bones Stirrup in each ear
3. Irregular bones Vertebra posterior (back side) of the body
4. Long bones Femur or thigh bone

6. What will happen if?

(a) There are no joints in our body.

If there are no joints in our body, we will be standing like a tree without any movement, we can move only because of joints.

(b) There is no melanin pigment in our body.

We will not have protection from ultraviolet rays. Our skin will become whitish.

(c) Instead of 33 vertebrae in our body, we had one single and straight bone.

We cannot bend down at our will.

7. Draw diagrams.

(a) Types of joints.



Hinge joint



**Ball and
socket
joint**



**Gliding
joint**



Some types of joints

(b) Structure of skin

